

# Report about the concentration of radon in the air

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## Contents

|          |                                      |           |
|----------|--------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                  | <b>2</b>  |
| <b>2</b> | <b>Theory</b>                        | <b>2</b>  |
| <b>3</b> | <b>Setup</b>                         | <b>3</b>  |
| <b>4</b> | <b>Results/Interpretation</b>        | <b>4</b>  |
| 4.1      | Energy calibration . . . . .         | 4         |
| 4.2      | Efficiency of the detector . . . . . | 4         |
| 4.3      | Spectrum of the filter . . . . .     | 5         |
| 4.4      | Radon concentration . . . . .        | 6         |
| <b>5</b> | <b>Conclusion</b>                    | <b>7</b>  |
| <b>6</b> | <b>Appendix</b>                      | <b>8</b>  |
| <b>7</b> | <b>References</b>                    | <b>12</b> |

## 1 Introduction

With radon being a highly radioactive element, which diffuses from the ground into air, we would like to know its concentration in room air. Then we can better know, if the concentration impacts our health. Because radon is a gas, we have to measure the concentration of its decay product and reverse engineer the radon concentration. In this experiment we are doing exactly that based on lead and polonium.

## 2 Theory

The decay law

$$\frac{dN}{dt} = -\lambda N, \quad (1)$$

describes the time evolution of the number of radioactive nuclei  $N$ .  $\lambda = \frac{1}{\tau}$  is called the decay constant, where  $\tau$  is the lifetime of the material. For the  $i$ -th isotope in a decay chain, the rate of change of the number of nuclei is

$$\frac{dN_i}{dt} = \lambda_{i-1}N_{i-1} - \lambda_i N_i. \quad (2)$$

A secular equilibrium is established, because the lifetimes of the parent isotopes in the decay chains of  $^{232}\text{Th}$  and  $^{238}\text{U}$  are much longer than those of their decay products. This is expressed by

$$\lambda_i N_i = \lambda_j N_j \quad (3)$$

for any two isotopes  $i$  and  $j$  in the same decay chain. Further is

$$\lambda_{Rn} \cdot \rho_{Rn} = \lambda_{Pb} \cdot \rho_{Pb}. \quad (4)$$

The decay rate of polonium, which is part of the radon decay chain and collected by an air filter with efficiency  $\epsilon_f$ , is given by

$$\frac{dN_{Po}}{dt} = \epsilon_f \rho_{Po} \cdot V - \lambda_{Po} N_{Po}, \quad (5)$$

where  $V$  is the volume of air passing through the filter and  $\rho_{Po}$  is the concentration of this isotope in the air. This relation is used to calculate the decay rate of lead

$$\frac{dN_{Pb}}{dt} = \lambda_{Po} N_{Po} + \epsilon_f \rho_{Pb} V - \lambda_{Pb} N_{Pb}. \quad (6)$$

The concentration of lead is then determined with

$$N_\gamma = \rho_{Pb} \epsilon_f \epsilon_d R V$$

$$\begin{aligned} & [(e^{-\lambda_{Pb} t_1} - e^{-\lambda_{Pb} t_2}) \left( \frac{1}{\lambda_{Po}} + \frac{1}{\lambda_{Pb}} + \frac{\lambda_{Pb}}{\lambda_{Po}(\lambda_{Po} - \lambda_{Pb})} + \frac{\lambda_{Po} e^{-\lambda_{Pb} T_f}}{\lambda_{Pb}(\lambda_{Pb} - \lambda_{Po})} \right) + \\ & \left( \frac{\lambda_{Pb}(1 - e^{-\lambda_{Po} T_f})(e^{-\lambda_{Po} t_1} - e^{-\lambda_{Po} t_2})}{\lambda_{Po}(\lambda_{Po} - \lambda_{Pb})} \right)], \end{aligned} \quad (7)$$

where  $N_\gamma$  is the number of lead decays.  $N_\gamma$  can be calculated using equation (5). The remaining required quantities are the branching ratio  $R$  of the gamma transition, the start and stop times  $t_1$  and  $t_2$  after the filter was switched off. The filtering time  $T_f$ , and the detection efficiency  $\epsilon_d$  for gamma particles of this energy (cf. [4]).

The efficiency  $\epsilon_d$  is calculated with

$$\epsilon_d = \frac{N_{\text{measured}}}{N_{\text{expected}}}. \quad (8)$$

The expected number of decays can be resolved with

$$N_{\text{expected}} = A(t_h) \cdot t_{\text{live}} \cdot R, \quad (9)$$

where  $A$  is the activity. Lastly, the concentration of radon can be determined by using

$$\rho_{Rn} \cdot \lambda_{Rn} = \rho_{Pb} \cdot \lambda_{Pb}. \quad (10)$$

Due to the secular equilibrium this is a valid approach to calculate the radon concentration.

### 3 Setup

Using a HEPA filter, the daughter isotopes of radon are collected on filter paper for about 70 minutes. The paper is then folded into a compact geometry so it can be approximated as a point radiation source.

During this process a germanium detector (HPGe detector), used to detect gamma radiation, is calibrated with samples of  $^{133}\text{Ba}$  and  $^{60}\text{Co}$ . The gamma emission lines of these isotopes are known. Each calibration measurement lasts about 10-15 minutes.

The folded paper is then placed in front of the detector and measured for about 3-4 hours. The time between the end of filtration and the start of the measurement is recorded.

Finally, the background radiation is measured overnight. (cf. [4])

## 4 Results/Interpretation

### 4.1 Energy calibration

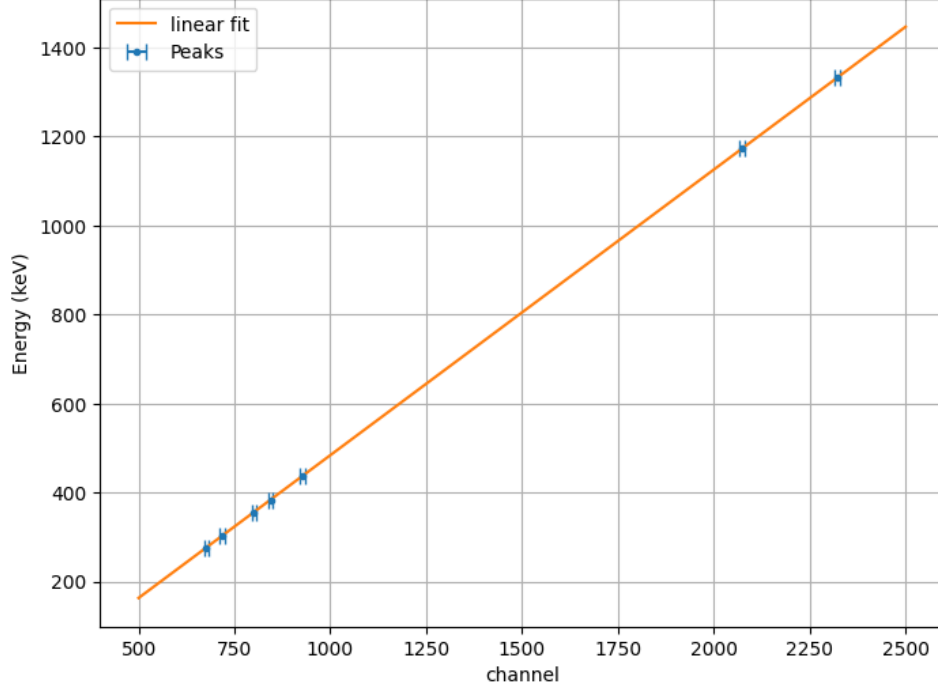


Figure 1: The energy peaks of the sample material in a channel-energy-diagram. The data is fitted with a linear curve

The peaks of the sample materials, which were derived from figure 4 and 5 are shown in fig. 1. After fitting the data, the energy can be calculated with  $E = 0,641(20) \text{ keV channel}^{-1} \cdot \text{channel} - 157,32(27) \text{ keV}$ .

### 4.2 Efficiency of the detector

The efficiency of the detector is plotted using the efficiency at each peak, calculated with equation (8). To compute the measured decays, a Gaussian curve was fitted to each peak in the spectrums of the calibration sources. All fitted parameters are listed in table 5. The resulting curve is shown in figure 2. All data points are well described with  $\epsilon_D(E) = 1.40(430) \cdot 10^{-5} \cdot E^{1.72(59)} \cdot e^{-4.20(980) \cdot 10^{-3} E}$ . This formula is later used for the determination of lead concentration.

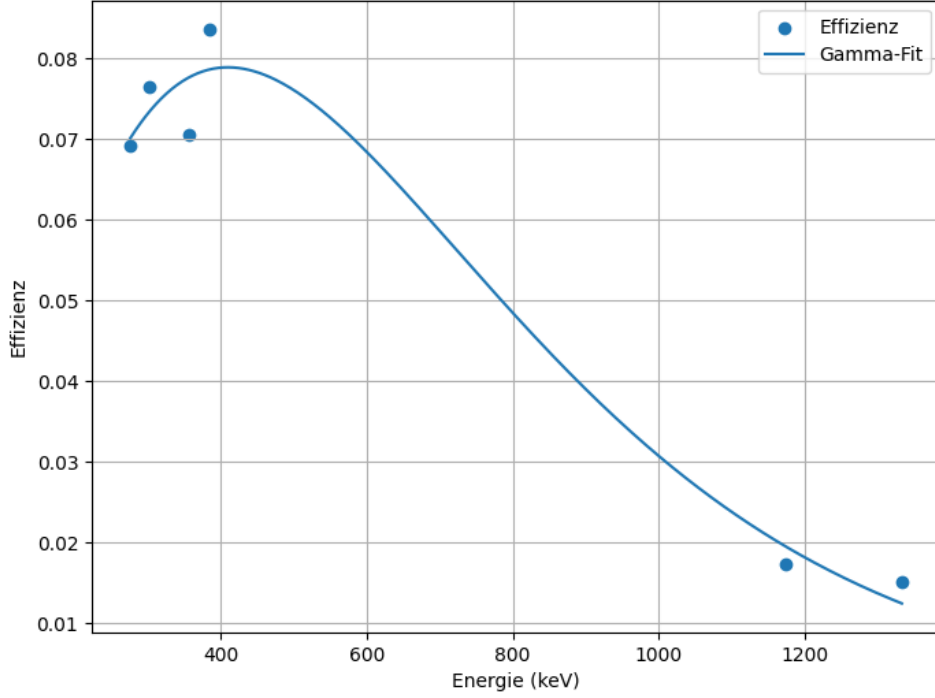


Figure 2: efficiency of the detector for each peak in an energy-efficiency-diagram. The data is fitted using a semi-empirical power-law function with exponential cutoff.

### 4.3 Spectrum of the filter

Figure 3 shows the measured spectrum of the filter paper and background radiation. The most visible photo peaks are marked. These correspond to  $^{214}\text{Pb}$ , the 510 keV peak of  $^{208}\text{Tl}$  and the 609keV peak of  $^{214}$ . Unfortunately the polonium peaks were not clearly visible in our measurement.

The filter efficiency is approximated by  $\epsilon_f \approx 1$ . This is justified since the usual efficiency for HEPA filters is approximately 99.97% and exact data for the used filter was not available (cf. [3]).

In the following section, the half lifes in tab. 4 are assumed. For the branching ratios and other properties the values shown in figure 6 and figure 7 are used. Using the Gaussian curves in table 6 for the area determination under the specified peaks, we calculated the gross and net rate of decays, efficiency and energy of the isotope peaks. The calculated results are shown in table 1.

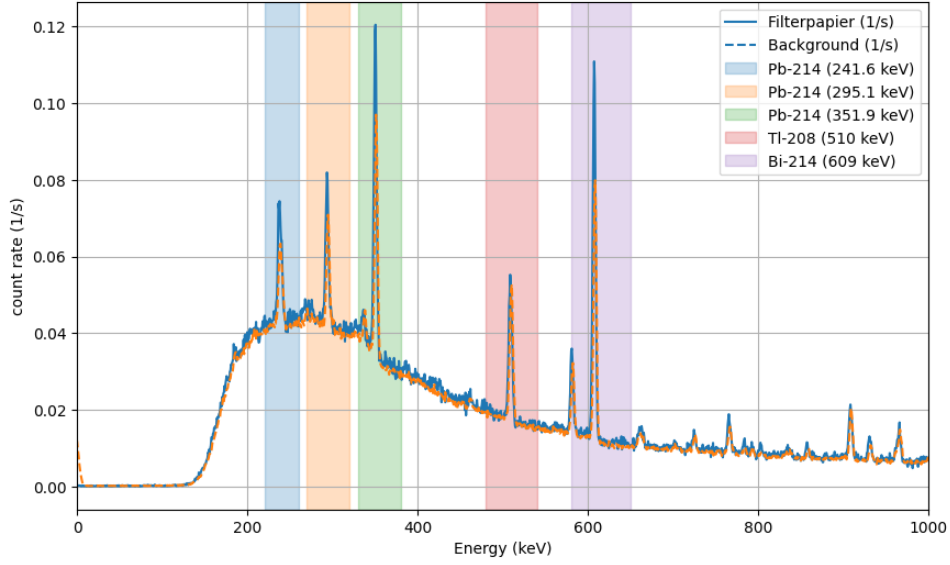


Figure 3: spectrum of the filter paper and background in a energy-rate-diagram

| Literature peak               | energy in keV | efficiency | gross rate in $s^{-1}$ | net rate in $s^{-1}$ |
|-------------------------------|---------------|------------|------------------------|----------------------|
| $^{214}\text{Pb}$ (241,6 keV) | 237(12)       | 0,063(28)  | 2,983(15)              | 0,167(16)            |
| $^{214}\text{Pb}$ (295,1 keV) | 293(14)       | 0,072(33)  | 3,588(16)              | 0,144(18)            |
| $^{214}\text{Pb}$ (351,9 keV) | 349(16)       | 0,077(35)  | 3,250(16)              | 0,210(17)            |
| $^{208}\text{Tl}$ (510 keV)   | 508(21)       | 0,075(36)  | 1,906(19)              | 0,055(13)            |
| $^{214}\text{Bi}$ (609 keV)   | 607(24)       | 0,067(33)  | 2,055(12)              | 0,244(13)            |

Table 1: Measurements for the rates, efficiency and energy of the peaks the  $^{220}\text{Rn}$  and the  $^{222}\text{Rn}$  decay chain, which could be clearly identified

#### 4.4 Radon concentration

Using equation 10 we now determine the concentration of radon. We can calculate it for each Energy Peak of lead. The individual results can be seen in 2. The weighted mean is  $\rho_{Rn} = 1.9(13)\text{Bq/m}^3$ .

| Measurement                   | $c_A(^{222}\text{Rn})$ |
|-------------------------------|------------------------|
| $^{214}\text{Pb}$ (241,6 keV) | 7.3(32)                |
| $^{214}\text{Pb}$ (295,1 keV) | 21.6(99)               |
| $^{214}\text{Pb}$ (351,9 keV) | 1.53(70)               |
| Weighted Mean                 | 1.9(13)                |

Table 2: Concentration of particles  $c_A(^{222}\text{Rn})$  in  $\text{Bq/m}^{-3}$

## 5 Conclusion

The typical radon concentration in room air in Germany ranges from 3 to 30 Bq/m<sup>3</sup> and the maximal legally permitted concentration is 300 Bq/m<sup>3</sup> (cf. [1]). Depending on the location in Germany the concentration can vary. In Garching it is approximately 52 Bq/m<sup>3</sup> (cf. [2]).

Our result of 1,9(13) Bq/m<sup>3</sup> is about one order of magnitude off of the expected value for Germany. Because we approximated the filter paper as a point source and neglected the solid angle of the detector, the calculation is not fully accurate. The fact that the liquid nitrogen used to cool the detector was not completely filled, might have resulted in an offset in the temperature. Due to the build of a germanium detector this could lead to slightly wrong detection rates. Since the room is frequently used for this specific FoPra it is filtered regularly. Therefore, the concentration may differ from that in a typical room in Garching. In our measurement it shows that the 351.9 keV is the most responsible energy peak for the radon concentration.

The result of this experiment could be improved by changing the location of the experiment every time, but this is rather complicated and impractical. Since only the peaks of <sup>214</sup>Pb were clearly identifiable, only the concentration of <sup>222</sup>Rn could be calculated.

## 6 Appendix

| Quantity                                            | Value                |
|-----------------------------------------------------|----------------------|
| Air volume pumped $V$                               | 56,05 m <sup>3</sup> |
| Pump duration $T_f$                                 | 4212 s               |
| Measurement time (filter paper) $t_{\text{filter}}$ | 13 485 s             |
| Delay time after filtration $t_1$                   | 58,61 s              |
| Background measurement time $t_{\text{bg}}$         | 66 400 s             |

Table 3: Experimental measurement parameters

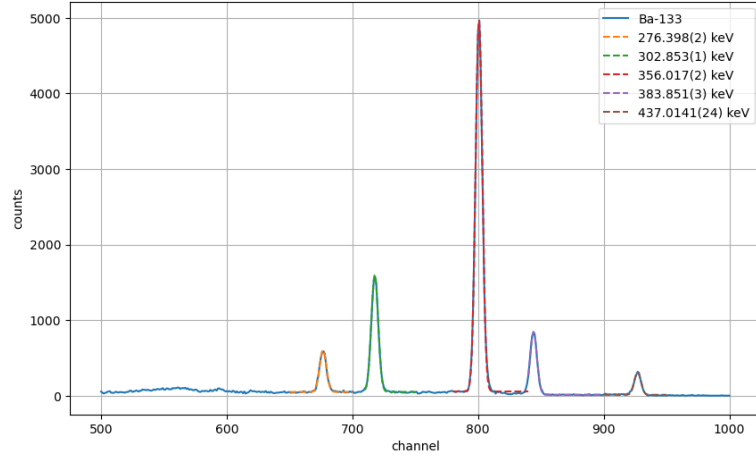


Figure 4: Spectrum of <sup>133</sup>Ba in a channel-counts-diagram. The spectrum was measured over time of 766,101 s



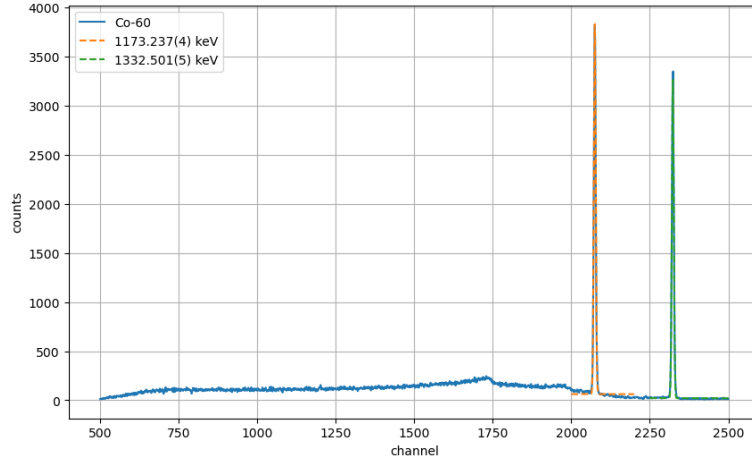


Figure 5: Spectrum of  $^{60}\text{Co}$  in a channel-counts-diagram. The spectrum was measured over a time of 759,369 s

| Material          | Half life time |
|-------------------|----------------|
| $^{60}\text{Co}$  | 5,2714(5) a    |
| $^{133}\text{Ba}$ | 10,51(5) a     |
| $^{216}\text{Po}$ | 0,15 s         |
| $^{218}\text{Po}$ | 3,11 min       |
| $^{214}\text{Pb}$ | 26,8 min       |
| $^{212}\text{Pb}$ | 10,64 h        |
| $^{220}\text{Rn}$ | 55,6 s         |
| $^{222}\text{Rn}$ | 3,82 d         |

Table 4: Half-life times of samples and most important members of the  $^{220}\text{Rn}$  and the  $^{222}\text{Rn}$  decay chain

| Energy Peak in keV                                          | A          | $\mu$        | $\sigma$   | c         |
|-------------------------------------------------------------|------------|--------------|------------|-----------|
| $^{60}\text{Co}$ 1173.237(4)                                | 3770(16)   | 2074.786(15) | 4.7169(48) | 62.5(23)  |
| $^{60}\text{Co}$ 1332.501(5)                                | 3256(9)    | 2323.26(99)  | 4.818(31)  | 21.6(11)  |
| $^{133}\text{Ba}$ 276.398(2)                                | 551.3(61)  | 676.573(33)  | 3.663(97)  | 48.3(17)  |
| $^{133}\text{Ba}$ 302.853(1)                                | 1543.2(63) | 717.798(12)  | 3.730(37)  | 49.7(21)  |
| $^{133}\text{Ba}$ 356.017(2)                                | 4903(23)   | 800.701(14)  | 3.610(41)  | 55.7(60)  |
| $^{133}\text{Ba}$ 383.851(3)                                | 833.3(39)  | 844.142(14)  | 3.642(43)  | 11.48(95) |
| $^{133}\text{Ba}$ 80.9971(14) + 356.017(2)<br>(Coincidence) | 299.5(48)  | 926.968(47)  | 3.44(13)   | 10.7(13)  |

Table 5: Values for the Gaussian fit  $A \cdot \exp(-\frac{(x-\mu)^2}{2\sigma^2}) + c$

| Energy peak in keV           | $A$      | $\mu$     | $\sigma$  | $m$       | $c$       |
|------------------------------|----------|-----------|-----------|-----------|-----------|
| $^{214}\text{Pb}$ 237.1(123) | 417(13)  | 238.17(8) | 2.20(8)   | 0.36(28)  | 495(66)   |
| $^{214}\text{Pb}$ 292.7(140) | 510(11)  | 293.74(4) | 1.71(4)   | -1.55(15) | 1036(45)  |
| $^{214}\text{Pb}$ 349.3(158) | 1129(16) | 350.40(2) | 1.544(25) | -3.91(21) | 1867(74)  |
| $^{208}\text{Tl}$ 508.3(208) | 499(7)   | 509.11(3) | 1.94(3)   | -0.99(8)  | 743(40)   |
| $^{214}\text{Bi}$ 606.8(238) | 1285(29) | 607.50(4) | 1.60(4)   | -1.87(24) | 1332(148) |

Table 6: Values for the Gaussian Fit  $A \cdot \exp(-\frac{(E-\mu)^2}{2\sigma^2}) + mE + c$

| Isotope           | Half-life               | Decay Gamma-ray Energy (keV) | Branching Ratio ( % ) | Decay Mode |       |
|-------------------|-------------------------|------------------------------|-----------------------|------------|-------|
| $^{220}\text{Rn}$ | 55.6 s                  | 549.7                        | 0.095                 | $\alpha$   | 100%  |
| $^{216}\text{Po}$ | 0.15 s                  | 804.9                        | 0.0018                | $\alpha$   | 100%  |
| $^{212}\text{Pb}$ | 10.64 hr                | 77.1                         | 17.5                  | $\beta^-$  | 100%  |
|                   |                         | 238.6                        | 43.6                  |            |       |
|                   |                         | 300.1                        | 3.34                  |            |       |
| $^{212}\text{Bi}$ | 60.55 min               | 727.2                        | 6.65                  | $\beta^-$  | 64.1% |
|                   |                         | 785.4                        | 1.11                  |            |       |
| $^{212}\text{Bi}$ | 60.55 min               | 1620.6                       | 1.51                  | $\alpha$   | 35.9% |
|                   |                         | 39.9                         | 1.10                  |            |       |
| $^{212}\text{Po}$ | $2.98 \times 10^{-7}$ s |                              |                       | $\alpha$   | 100%  |
| $^{208}\text{Tl}$ | 3.05 min                | 72.8                         | 2.03                  | $\beta^-$  | 100%  |
|                   |                         | 75.0                         | 3.43                  |            |       |
|                   |                         | 84.8                         | 1.18                  |            |       |
|                   |                         | 277.4                        | 6.80                  |            |       |
|                   |                         | 510.8                        | 21.6                  |            |       |
|                   |                         | 583.1                        | 86.0                  |            |       |
|                   |                         | 763.1                        | 1.64                  |            |       |
|                   |                         | 860.4                        | 12.0                  |            |       |
|                   |                         | 2614.6                       | 99.8                  |            |       |
| $^{208}\text{Pb}$ | stable                  |                              |                       |            |       |

Figure 6: properties of isotopes in the  $^{220}\text{Rn}$  decay chain

| Isotope           | Half-life               | Decay Gamma-ray Energy (keV)                                                                                                                                                                    | Branching Ratio ( % )                                                                                                                                        | Decay Mode                         |
|-------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <sup>222</sup> Rn | 3.82 d                  |                                                                                                                                                                                                 |                                                                                                                                                              | $\alpha$ 100%                      |
| <sup>218</sup> Po | 3.11 min                | 837.0                                                                                                                                                                                           | 0.0011                                                                                                                                                       | $\alpha$ 99.98%<br>$\beta^-$ 0.02% |
| <sup>214</sup> Pb | 26.8 min                | 77.1<br>241.9<br>295.2<br>351.9<br>785.9                                                                                                                                                        | 10.8<br>7.46<br>19.2<br>37.1<br>1.09                                                                                                                         | $\beta^-$ 100%                     |
| <sup>214</sup> Bi | 19.9 min                | 609.3<br>665.5<br>768.4<br>806.2<br>934.1<br>1120.3<br>1155.2<br>1238.1<br>1281.0<br>1377.7<br>1401.5<br>1408.0<br>1509.2<br>1661.3<br>1729.6<br>1764.5<br>1847.4<br>2118.5<br>2204.1<br>2447.7 | 46.1<br>1.56<br>4.88<br>1.23<br>3.16<br>15.0<br>1.69<br>5.92<br>1.47<br>4.02<br>1.39<br>2.48<br>2.19<br>1.15<br>3.05<br>15.9<br>2.12<br>1.21<br>4.99<br>1.55 | $\beta^-$ 99.98%<br>$\alpha$ 0.02% |
| <sup>214</sup> Po | $1.64 \times 10^{-4}$ s | 799.7                                                                                                                                                                                           | 0.0104                                                                                                                                                       | $\alpha$ 100%                      |
| <sup>210</sup> Tl | 1.3 min                 | 298<br>797.7<br>860<br>1070<br>1110<br>1210<br>1316<br>2010<br>2360<br>2430                                                                                                                     | 80<br>100<br>7<br>12<br>7<br>17<br>21<br>7<br>8<br>9                                                                                                         | $\beta^-$ 100%                     |

Figure 7: properties of isotopes in the <sup>222</sup>Rn decay chain

## 7 References

### References

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